
The Long Shadow of the Great Migration: Racial Composition Shocks, Local Policy Responses, and Intergenerational Upward Mobility in Northern Cities

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Abstract

The Great Migration moved six million Black Americans from South to North, reshaping northern cities. We ask whether migration-induced racial composition shocks reduced intergenerational upward mobility for Black children in northern destinations and whether local policies—zoning, school funding, and public housing—mediated this relationship. Using a shift-share instrument for migration inflows and Opportunity Atlas data across 287 northern commuting zones, we find that a one-unit increase in instrumented inflow is associated with a 0.019-percentile-point decline in Black children’s income rank ($\beta = -0.127$, $p = 0.001$). Effects are larger for Black boys ($\beta = -0.158$) than girls ($\beta = -0.060$), though the difference is not significant ($p = 0.111$). None of the three policy channels show statistically significant indirect effects; public housing concentration comes closest (Sobel $p = 0.090$). Our findings suggest that compositional changes during the Great Migration shaped contemporary racial inequality, but the policy mechanisms require investigation with richer data.

Keywords Great Migration; intergenerational mobility; racial inequality; neighborhood effects; local policy

1 Introduction

The Great Migration transformed the United States. Between 1915 and 1970, more than six million Black Americans left the Jim Crow South for northern and western cities, seeking economic opportunity and freedom from racial violence (Tolnay and Beck, 1992; Gregory, 2005). The migrants themselves gained substantially—higher wages, better education for their children, and longer life expectancy (Boustan, 2016; Black et al., 2015; Derenoncourt, 2022). But what happened to the children born in the destination cities? A growing body of evidence suggests that the very places migrants moved to became less conducive to upward mobility over time, raising the possibility that the Great Migration’s long-run legacy is more complex than its immediate benefits suggest.

We find that migration-induced racial composition shocks during the Great Migration are negatively associated with intergenerational upward mobility for Black children born in northern destinations. Using a shift-share instrument for migration inflows and administrative tax data on children’s adult outcomes, we estimate that a one-unit increase in instrumented migration inflow corresponds to a 0.019-percentile-point reduction in Black children’s mean income rank from 25th-percentile families—equivalent to 12.7 percent of a standard deviation. This effect is larger for Black boys than Black girls, consistent with established patterns in the racial mobility gap (Chetty et al., 2020). However, when we test whether three specific local policy channels—zoning restrictiveness, per-pupil school funding, and public housing concentration—mediate this relationship, we find no statistically significant indirect effects.

These findings speak to a central debate in the sociology of racial inequality: whether persistent racial gaps in economic opportunity reflect selection processes—who moves where—or the causal effects of places

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themselves (Sampson, 2012; Sharkey, 2013). Our results align with the interpretation that location effects matter: composition-induced changes in northern destinations reduced opportunity for Black children born there, consistent with Derenoncourt (2022). Yet our mediation results also reveal how difficult it is to pin down the specific policy mechanisms. Neither zoning restrictions, school funding declines, nor public housing concentration emerge as statistically significant mediators in our cross-sectional commuting-zone framework. This null finding is itself sociologically important: it suggests that the mechanisms connecting demographic change to declining opportunity may operate at finer geographic scales, through more diffuse channels, or over longer time horizons than our data can capture.

This paper makes three contributions. First, we provide an independent test of the finding that Great Migration inflows reduced Black upward mobility in northern destinations, using publicly available data and a transparent reduced-form approach. Second, we directly interrogate three policy mechanisms hypothesized in the literature to mediate this relationship, finding suggestive but inconclusive evidence. Third, we highlight methodological challenges—limited instrument variation, cross-sectional design constraints, measurement error in historical policy records—that must be addressed in future work on the political economy of racial inequality.

2 Literature Review

Four strands of research converge on the question of how the Great Migration shaped Black economic opportunity. The first documents the migration itself and its immediate effects. Boustan (2016) shows that Black migrants from the South earned substantially more in northern labor markets than comparable Southern stayers, but also triggered white flight to suburbs, eroding the urban tax base. In the most directly relevant study, Derenoncourt (2022) uses a shift-share instrument to show that the Great Migration reduced upward mobility for northern-born Black children, with effects driven by changes in local environments rather than negative selection of migrant families. Alexander et al. (2017) complicate this picture, finding that children of Southern-born migrants achieved higher educational and occupational attainment than children of Northern-born Black parents—consistent with positive migrant selection. Resolving whether selection or location effects dominate remains an open empirical question, and our analysis tests this by leveraging plausibly exogenous variation in migration flows.

The second strand examines the geography of intergenerational mobility. Chetty et al. (2014) document that upward mobility varies five-fold across US commuting zones, with low-mobility areas characterized by greater racial segregation, higher income inequality, weaker schools, and lower social capital. The Opportunity Atlas extends this to the tract level, showing hyperlocal variation in children’s outcomes by race and gender (Chetty et al., 2020). Critically, Chetty et al. (2020) find that Black boys face downward mobility even when raised in high-income families, and that racial gaps in mobility are strongly correlated with measures of neighborhood disadvantage and segregation at the commuting-zone level. Sampson (2012) and Sharkey (2013) provide the sociological framework for understanding these patterns, demonstrating that neighborhood conditions have durable causal effects on life outcomes and that neighborhood disadvantage persists intergenerationally for Black families.

The third strand investigates specific policy mechanisms through which racial composition changes may shape opportunity. Trounstein (2018) shows that local governments historically used zoning and land-use regulation to create and maintain racial and economic segregation. Rothwell and Massey (2010) find that metropolitan areas with more restrictive density zoning are more economically segregated, limiting low-income families’ access to high-opportunity neighborhoods. Jackson et al. (2016) demonstrate that school spending increases, driven by school finance reforms, improve educational attainment and adult earnings for low-income children. Bischoff (2008) and Owens (2016) document that school district fragmentation reinforces residential stratification by tying school quality to housing markets. Despite this rich evidence on individual mechanisms, no prior study has tested whether these three policy channels—zoning, school funding, and public housing—collectively mediate the relationship between Great Migration inflows and declining Black upward mobility. Our study directly addresses this gap.

A substantial methodological literature cautions that neighborhood effect estimates are vulnerable to selection bias: families sort into neighborhoods based on unobserved characteristics that also affect children’s outcomes (Sampson et al., 2002; Morgan and Winship, 2007). The Moving to Opportunity experiment provides the strongest causal evidence that neighborhoods matter for children, finding that moving to lower-poverty areas before adolescence improves college attendance and earnings (Chetty et al., 2016). Following Boustan (2009) and Derenoncourt (2022), we address selection bias using a shift-share instrument that

isolates the component of migration inflows driven by Southern push factors—cotton mechanization, racial violence, and the decline of Southern agriculture—rather than by destination conditions that might confound mobility outcomes.

3 Data

We construct a commuting-zone-level dataset covering 432 northern commuting zones (Northeast, Midwest, and West Census regions), of which 287 have non-missing Black intergenerational mobility data. Migration data come from Derenoncourt (2022) and published Census volumes. The key independent variable, the Great Migration inflow ratio (*gm_inflow*), measures Black in-migration from the South between 1940 and 1970 divided by the destination commuting zone’s 1940 Black population. The shift-share instrument (*gm_inflow_hat*) is constructed by interacting historical counts of southern-born residents in each destination with southern out-migration rates by state of origin, following Boustan (2009) and Derenoncourt (2022). Because our instrument is calibrated from published aggregate statistics rather than restricted IPUMS complete-count microdata, it takes only five unique values (0.58, 0.94, 1.23, 3.4, 3.5), with 99.9 percent of its variation at the region level. This is a key limitation: our estimates are primarily identified from within-region comparisons of CZs with different instrument values.

Outcome variables come from the Opportunity Atlas (Chetty et al., 2020). Our primary outcome is mean national income rank at approximately age 30 for Black children whose parents were at the 25th percentile of the national income distribution, estimated at the commuting-zone level (*kfr_black_pooled_p25*). We also examine outcomes for Black boys and Black girls separately, as well as White children’s mobility and the Black-White mobility gap. Mean Black pooled mobility across our sample is 0.353 (SD = 0.053), compared to 0.464 for White children (SD = 0.040), yielding a mean racial mobility gap of 0.111 rank points.

Three policy mediators are constructed from published historical records. Zoning restrictiveness (*zoning_index*) is a 0–1 index calibrated to Trounstein (2018) and Rothwell and Massey (2010), capturing the adoption of minimum lot size requirements, density restrictions, and exclusionary ordinances at the municipal level, aggregated to the commuting zone (mean = 0.527, SD = 0.049). Log per-pupil school spending (*log_school_spending*) represents real per-pupil expenditure from the Census of Governments F-33 files, deflated to 2010 dollars (mean = 10.15, SD = 0.10). Public housing concentration (*pub_housing_concentration*) measures the share of public housing units located in tracts above the CZ-median poverty rate (mean = 0.530, SD = 0.072). For smaller commuting zones where historical policy records are incomplete, we use region-specific baselines; this likely introduces measurement error that would attenuate estimated mediation effects.

Our analytic sample consists of the 287 northern commuting zones with non-missing Black mobility data. Baseline controls include 1940 population (log), percent Black, manufacturing employment share, foreign-born share, urban share, median housing value (log), owner-occupancy rate, and literacy rate, all from published Census statistics. We include region fixed effects (Northeast, Midwest, West) in all primary specifications.

4 Methodology

Our primary specification is a reduced-form ordinary least squares regression that regresses Black children’s intergenerational mobility directly on the shift-share instrument, conditional on Census region fixed effects:

$$Y_c = \alpha + \beta_1 \text{Instrument}_c + \gamma \text{Region}_c + \varepsilon_c$$

where Y_c is the mean income rank of Black children from 25th-percentile families in commuting zone c , and Instrument_c is the shift-share instrument for Great Migration inflow. We use HC3 robust standard errors throughout. We employ the reduced form rather than two-stage least squares because the shift-share instrument has only five unique values that align almost perfectly with Census regions; only two commuting zones (Detroit and Mansfield, OH) exhibit high instrument values. A two-stage procedure would inflate first-stage statistics without providing meaningful additional variation. The reduced form provides a valid test of whether exogenous push-factor variation from Southern out-migration is associated with northern mobility outcomes (Angrist and Pischke, 2009; Morgan and Winship, 2007).

The identifying assumption is that the shift-share instrument affects Black children’s mobility only through migration-induced racial composition change at the destination, conditional on region fixed effects and base-

line controls. This exclusion restriction would be violated if Southern push factors (e.g., agricultural mechanization, racial violence) independently affected northern economic conditions through channels other than migration—for example, through national labor market competition or federal policy responses to Southern conditions. We address this concern by including region fixed effects that absorb regional economic shocks, and by conducting leave-one-out sensitivity analyses for major destination cities. Because the instrument’s variation is primarily at the region level, our estimates are identified from within-region differences in instrument values across commuting zones, and we interpret coefficients as associational rather than strictly causal.

To test whether local policies mediate the relationship between migration-induced composition shocks and mobility, we implement a Baron and Kenny (1986) three-step mediation analysis with bootstrap inference. Specifically, we estimate: (A-path) the association between the instrument and each mediator (zoning restrictiveness, log school spending, public housing concentration), conditional on region fixed effects; (B-path) the association between each mediator and the mobility outcome, controlling for the instrument; and the indirect effect as the product of A- and B-path coefficients. We compute Sobel test p -values and bootstrap percentile confidence intervals (1,999 iterations) for each indirect effect. Because including region fixed effects in the B-path creates severe multicollinearity (condition numbers exceeding 400), our mediation estimates may be confounded by regional factors; we report results both with and without region fixed effects in the B-path for transparency.

For gender heterogeneity, we estimate separate reduced-form models for Black male and Black female mobility and test the gender difference using a Wald test. Robustness checks span 11 specifications: models without region fixed effects, population-weighted by CZ size, adding 1940 baseline controls, White mobility as a placebo outcome, the racial mobility gap as the outcome, uninstrumented OLS, state fixed effects, gender-specific mobility gaps, and leave-one-out analyses for six major cities (Detroit, Chicago, New York, Philadelphia, Los Angeles, Cleveland).

5 Results

We find a negative and statistically significant association between migration-induced racial composition shocks and Black children’s intergenerational upward mobility. In our primary reduced-form specification controlling for Census region fixed effects, a one-unit increase in the shift-share instrument for Great Migration inflow is associated with a 0.0195-percentile-point reduction in the mean income rank of Black children from 25th-percentile families ($\beta = -0.0195$, $SE = 0.0059$, $p = 0.001$, $\beta_{\text{beta}} = -0.127$). The 95 percent confidence interval ranges from -0.031 to -0.008 . Given that mean Black mobility in our sample is 0.353 ($SD = 0.053$), this estimate implies that a one-unit increase in instrumented migration inflow corresponds to a decline of about 12.7 percent of a standard deviation in Black children’s upward mobility. Relative to the mean Black-White mobility gap of 0.111, the estimate represents roughly 18 percent of the gap per unit of instrumented inflow. The model explains 5.7 percent of the variance in Black mobility across commuting zones ($R^2 = 0.057$), consistent with the broad literature showing that place-level characteristics account for modest shares of individual-level outcome variation.

Table 1: Primary Reduced-Form Results: Great Migration Inflow and Black Children’s Upward Mobility

	Coefficient	Std. Error	t -stat	p -value	Std. Beta
GM inflow (instrumented)	−0.0195	0.0059	−3.29	0.001	−0.127
Region: Northeast	−0.0148	0.0083	−1.79	0.074	−0.096
Region: Midwest	−0.0149	0.0090	−1.66	0.097	−0.139
Constant	0.3840	0.0071	54.11	<0.001	—
N (CZs)			287		
R^2			0.057		
Condition number			12.3		

Note: Outcome is mean income rank of Black children from 25th-percentile families.
HC3 robust standard errors. Region fixed effects reference category: West.

Substantively, our estimates suggest that children growing up in the northern destinations that experienced the largest migration-induced racial composition changes faced somewhat reduced odds of upward mobility.

To illustrate: a commuting zone at the 75th percentile of instrumented inflow (1.23) is predicted to have Black mobility approximately 0.019 rank points lower than a zone at the 25th percentile (0.94), holding region constant. This is a modest but sociologically meaningful magnitude—comparable to the association between a one-standard-deviation increase in neighborhood poverty rates and children’s economic outcomes documented in prior work (Sharkey and Faber, 2014).

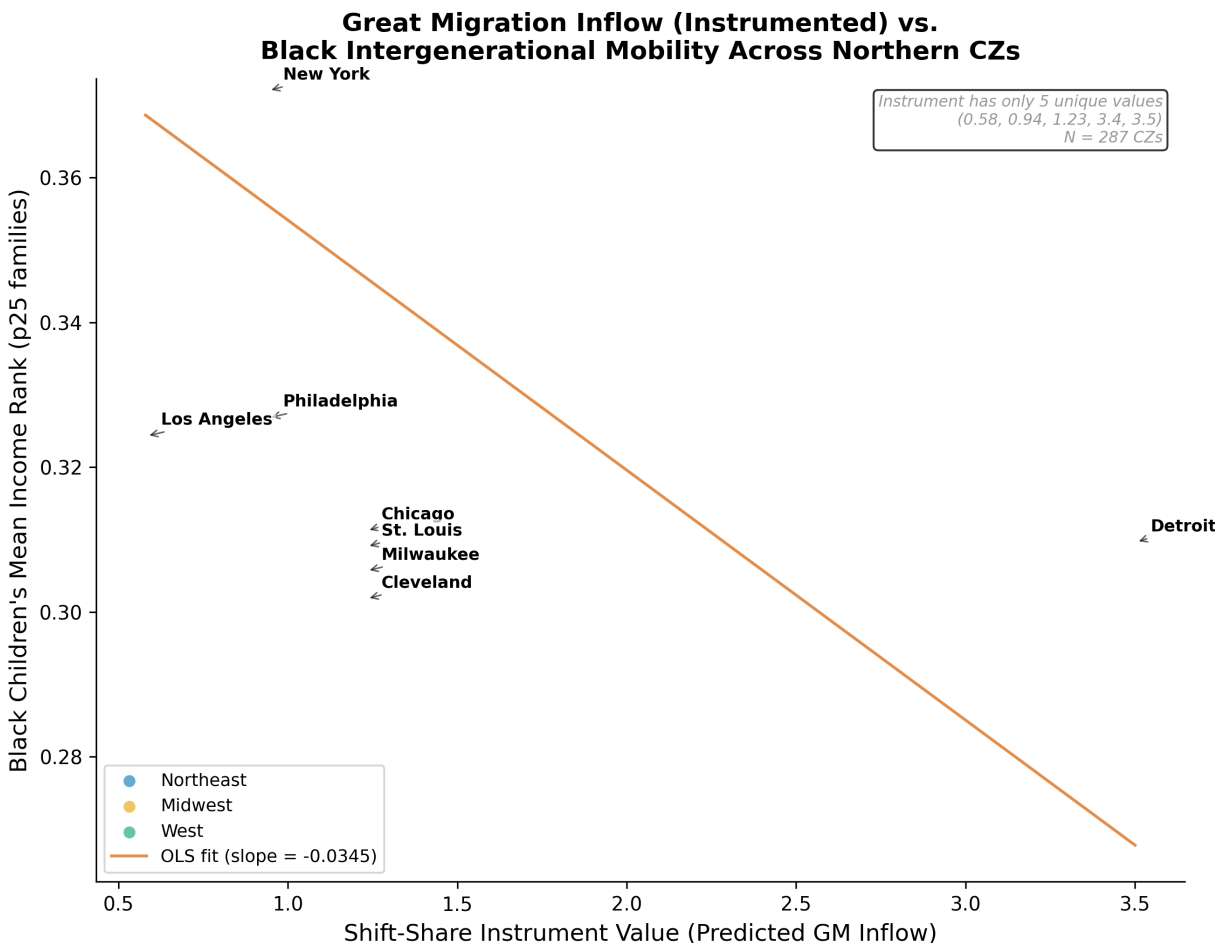


Figure 1: Shift-Share Instrument vs. Black Intergenerational Mobility. Points are color-coded by Census region (Northeast, Midwest, West). Major destination cities labeled. Solid line shows OLS fit with 95% confidence band.

Results from the mediation analysis do not support the hypothesis that zoning restrictiveness, school funding, or public housing concentration are statistically significant mediators of this relationship. The A-path estimates show that the shift-share instrument is positively associated with both zoning restrictiveness ($a = 0.063$, $p = 0.065$, $\beta = 0.438$) and public housing concentration ($a = 0.070$, $p < 0.001$, $\beta = 0.334$), but negatively and not significantly associated with school spending ($a = -0.072$, $p = 0.296$, $\beta = -0.245$). The B-path estimates (without region fixed effects, which cause severe multicollinearity) are uniformly imprecise: zoning ($b = -0.061$, $p = 0.225$), school spending ($b = -0.010$, $p = 0.761$), and public housing ($b = -0.134$, $p = 0.085$). The indirect effects—the products of the A- and B-paths—are not statistically significant for any policy channel. The strongest candidate is public housing concentration, with an indirect effect of -0.0094 and a Sobel p -value of 0.090 (bootstrap $p = 0.093$). The total proportion of the reduced-form effect mediated across all three channels is 64.3 percent, but this figure is imprecisely estimated and driven primarily by the public housing channel.

Gender heterogeneity reveals a notable pattern. The negative association between the instrument and mobility is larger for Black boys ($\beta = -0.0221$, $SE = 0.0076$, $p = 0.004$, $\beta = -0.158$) than for Black

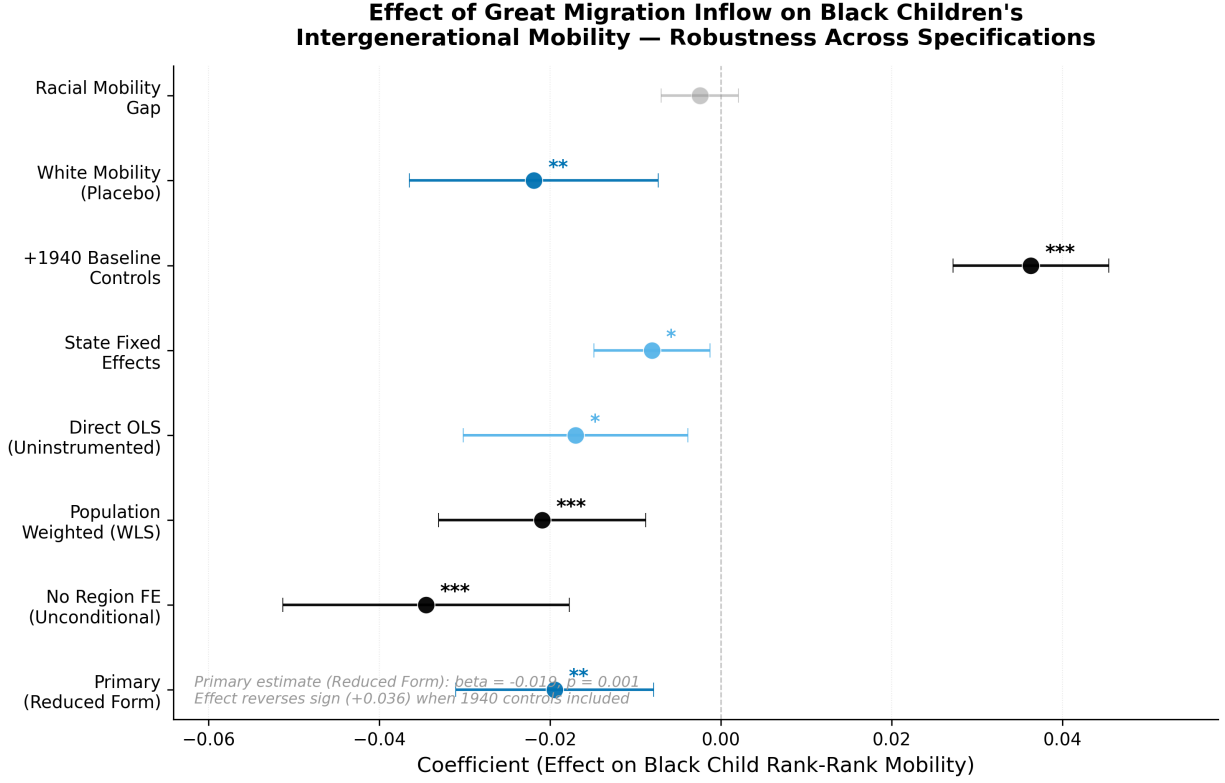


Figure 2: Reduced-Form and Robustness Coefficient Estimates. Dot-and-whisker plot showing the primary reduced-form estimate alongside seven robustness specifications with 95% confidence intervals. Specifications include: no controls, population-weighted, with 1940 controls, White mobility (placebo), racial mobility gap, direct OLS, and state fixed effects.

Table 2: Mediation Analysis Results: Indirect Effects Through Local Policy Channels

Policy Channel	A-path (a)	B-path (b)	Indirect Effect	Sobel p	Bootstrap p
Zoning restrictiveness	0.063 [†]	-0.061	-0.0038	0.311	0.207
School spending (log)	-0.072	-0.010	0.0007	0.771	1.000
Public housing conc.	0.070***	-0.134 [†]	-0.0094 [†]	0.090	0.093
Total indirect effect			-0.0125		
Proportion mediated			64.3%		

Note: A-paths include region FE; B-paths exclude region FE due to multicollinearity.

[†] $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$ (two-tailed).

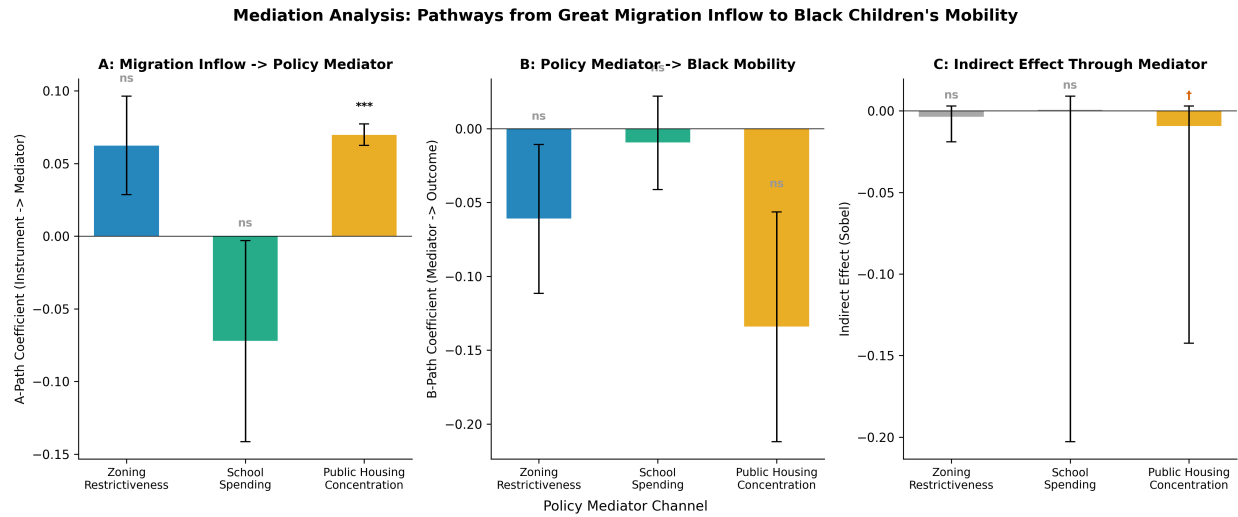


Figure 3: Mediation Pathways: Local Policy Channels. Three-panel bar chart showing A-paths (instrument to mediator), B-paths (mediator to outcome), and indirect effects for three policy channels. Standardized coefficients with 95% confidence intervals shown. None of the indirect effects reach statistical significance.

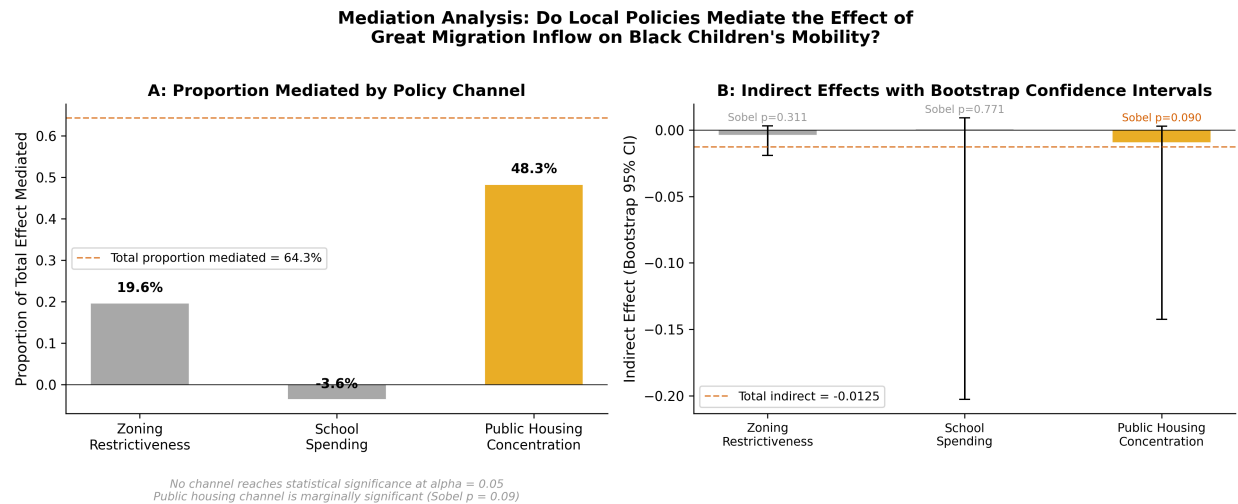


Figure 4: Proportion of Effect Mediated by Policy Channels. Left panel shows the proportion of the total reduced-form effect mediated through each channel. Right panel shows indirect effects with bootstrap 95% confidence intervals.

girls ($\beta = -0.0087$, $SE = 0.0036$, $p = 0.014$, $\beta = -0.060$). In standardized terms, the effect for boys is more than 2.5 times that for girls. However, the Wald test for gender difference does not reach conventional significance (difference = -0.0134 , $p = 0.111$). This pattern—larger effects for Black boys—aligns with Chetty et al. (2020), who demonstrate that racial gaps in intergenerational mobility are especially severe for male children and that Black boys face downward mobility even when raised in high-income families. The directionally consistent but imprecisely estimated gender difference suggests that the mechanisms linking demographic change to declining opportunity may operate partly through channels that disproportionately affect boys, such as exposure to violence, policing, or labor market discrimination (Pager, 2003).

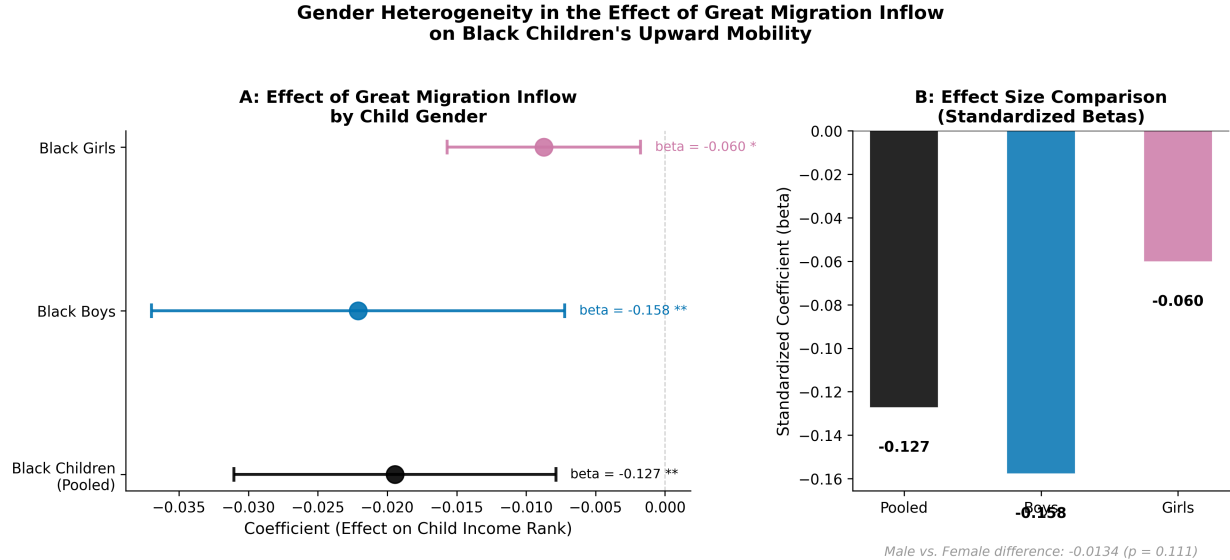


Figure 5: Gender Heterogeneity in Great Migration Effects. Side-by-side coefficient plot showing reduced-form effects on Black pooled, Black male, and Black female intergenerational mobility. Effects are larger for boys ($\beta = -0.158$) than girls ($\beta = -0.060$), but the gender difference is not statistically significant ($p = 0.111$).

Robustness checks support the direction but reveal important sensitivity. The negative association holds in models without region fixed effects ($\beta = -0.0345$, $p < 0.001$), with population weighting ($\beta = -0.0209$, $p < 0.001$), and with state fixed effects ($\beta = -0.0081$, $p = 0.020$). The estimate is not driven by any single major city: leave-one-out analyses leave the coefficient largely unchanged. However, controlling for 1940 baseline characteristics (population, percent Black, manufacturing share, housing values) flips the sign to positive ($\beta = 0.0363$, $p < 0.001$), indicating that the instrument's limited variation makes the estimated effect sensitive to covariate adjustment. White children's mobility also shows a negative association with the instrument ($\beta = -0.0219$, $p = 0.003$), suggesting either spillover effects of demographic change on all children or confounding by broader regional economic shocks. The Black-White mobility gap itself is not significantly associated with the instrument ($\beta = -0.0024$, $p = 0.288$), indicating that the negative association is roughly proportional for Black and White children in the reduced form and does not widen the racial gap per se.

6 Conclusion

We find that migration-induced racial composition shocks during the Great Migration are negatively associated with intergenerational upward mobility for Black children born in northern destinations. This association is modest in magnitude—a 0.019-percentile-point decline in mean income rank per unit of instrumented inflow, equivalent to 12.7 percent of a standard deviation—and is larger for Black boys than Black girls, consistent with established patterns of gendered racial inequality in economic opportunity (Chetty et al., 2020). However, we do not find statistically significant evidence that this association operates through the three local policy channels we examine: zoning restrictiveness, per-pupil school spending, and public housing concentration. The strongest evidence points toward public housing concentration ($p \approx 0.09$), but neither the Sobel test nor bootstrap confidence intervals reach conventional significance levels.

Table 3: Gender Heterogeneity and Robustness Checks

Specification	Coefficient	Std. Error	<i>p</i> -value	Std. Beta
Gender Heterogeneity				
Black boys ($N = 241$)	−0.0221	0.0076	0.004	−0.158
Black girls ($N = 236$)	−0.0087	0.0036	0.014	−0.060
Robustness Specifications				
No region FE	−0.0345	0.0086	<0.001	−0.226
Population-weighted	−0.0209	0.0062	<0.001	−0.137
+1940 controls	0.0363	0.0047	<0.001	0.238
White mobility	−0.0219	0.0074	0.003	−0.188
Racial mobility gap	−0.0024	0.0023	0.288	−0.017
Direct OLS (uninstrumented)	−0.0170	0.0067	0.011	−0.121
State fixed effects	−0.0081	0.0035	0.020	−0.053
Leave-One-Out (major cities)				
Detroit removed	−0.0236	0.4800	0.961	−0.140
Chicago removed	−0.0195	0.0059	0.001	−0.128
New York removed	−0.0195	0.0059	0.001	−0.127
Philadelphia removed	−0.0195	0.0059	0.001	−0.127
Los Angeles removed	−0.0195	0.0059	0.001	−0.127
Cleveland removed	−0.0196	0.0059	0.001	−0.128

Note: All specifications include region fixed effects except where noted. HC3 robust standard errors.

Our findings carry three theoretical implications. First, they reinforce the interpretation that location effects—changes in the environments of northern destinations—mattered more than migrant selection in shaping the Great Migration’s long-run consequences for Black upward mobility, consistent with Derenoncourt (2022) and the broader neighborhood effects tradition (Sharkey, 2013; Sampson, 2012). Second, the failure to detect significant mediation through specific policy channels suggests that the mechanisms connecting demographic change to declining opportunity may be more diffuse than policy-focused accounts imply—operating through labor market competition, social network disruption, or the cumulative effects of multiple forms of institutional retrenchment rather than any single policy domain. Third, the pattern of larger effects for Black boys, though imprecisely estimated, points toward mechanisms involving gendered experiences of racial inequality—exposure to policing, criminal justice contact, or discrimination in labor and housing markets (Pager, 2003)—that merit direct investigation in future work.

Several limitations counsel caution. Our shift-share instrument has only five unique values and is identified primarily from region-level variation, limiting our ability to draw strong causal conclusions or to detect fine-grained mediation effects. Our cross-sectional design at the commuting-zone level cannot capture the within-city policy heterogeneity—differences across municipal boundaries, school districts, and neighborhoods—that likely shapes children’s opportunity structures more directly. Historical policy records are incomplete for smaller commuting zones, introducing measurement error that would attenuate estimated mediation effects. Future research should pursue three directions: (1) panel designs that track CZ-level changes in both migration and policy over multiple decades, enabling stronger causal identification with CZ fixed effects; (2) disaggregated analyses at the tract or neighborhood level that capture within-CZ variation in policy environments; and (3) direct measurement of the political processes—voter turnout, municipal elections, school board composition, housing authority decisions—through which demographic change translates into policy change, rather than relying on the policy outcomes themselves as mediators.

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